

AI Audit Trail – HW06 API Testing

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Tool: Cursor AI Agent
Date range: 2026-08-28 – 2026-08-30

Interaction Log (12 entries)

#	Date	Prompt / Action	AI Output	Student Action
1	08-28	"Generate FR-02 login test matrix from api_specification.md"	30 positive/negative/security cases	Reviewed expected status codes against server.js
2	08-28	"Add boundary cases for email/password partitions"	FR02-D01-D10 domain cases	Kept 8, merged 2 duplicates
3	08-28	"Generate FR-08 checkout cases including cart preconditions"	30 checkout scenarios	Adjusted total_amount to match AirPods price
4	08-29	"Add security tests: SQLi, XSS, tampered total"	FR08-S01-S05	Confirmed SUT accepts tampered totals → INCOMPLETE label
5	08-29	"Generate FR-19 admin user management tests"	GET/DELETE matrix	Added IDOR cases after reading authenticateToken middleware
6	08-29	"Write build-postman-from-cases.js generator"	Node script with prerequest hooks	Tested locally, fixed async cart setup
7	08-29	"Add 5 student extension cases per pool"	STU01-STU05 per FR	Authored Vietnamese email, newline address, trailing slash tests
8	08-30	"Assign audit labels VALID/INVALID/INCOMPLETE/EXT"	Label mapping function	Manual review of 12 INCOMPLETE state-dependent cases
9	08-30	"Create GitHub Actions workflow with fail_one_test input"	api-tests.yml	Added workflow_dispatch choice input
10	08-30	"Generate ai-critique and self-assessment docs"	250-word critique draft	Expanded with specific bug references
11	08-30	"Fix admin password – seed uses Admin123! not Admin1234!"	Corrected environment file	Verified against database.js seed
12	08-30	"Run full Newman suite on port 3010"	Collection build + test execution	Recorded pass/fail in main-report

Authorship Split

Source	Cases	Percentage
AI-generated (reviewed)	~90	~86%
Student EXT (STU01-05 × 3)	15	~14%
Manual edits to expected values	~25	–

Verification Steps Taken

1. Cross-referenced every expected status with `eshop-sut/backend/server.js`
2. Ran `npm run build` after each case file change
3. Spot-checked 10 random cases in Postman GUI
4. Documented discrepancies in `docs/bugs.md`

Prompt Templates Used

```

Given API spec section [X], generate N test cases as JSON with fields:  
id, title, feature, method, endpoint, category, auditLabel,  
preconditions,  
headers, body, expectedStatus, expectedBodyContains, notes  
```

```

Convert JSON cases to Postman v2.1 collection with collection-level  
pre-request: X-Student-Id: 23127153  
```

Files Produced via AI

- `data/test-cases/\*.json`
- `scripts/build-postman-from-cases.js`
- `postman/23127153\_EShop\_API.postman\_collection.json` (generated)
- `docs/\*.md`
- `skills/hw06-api-test-generator/SKILL.md`
- `.github/workflows/api-tests.yml`